# Brass frame build

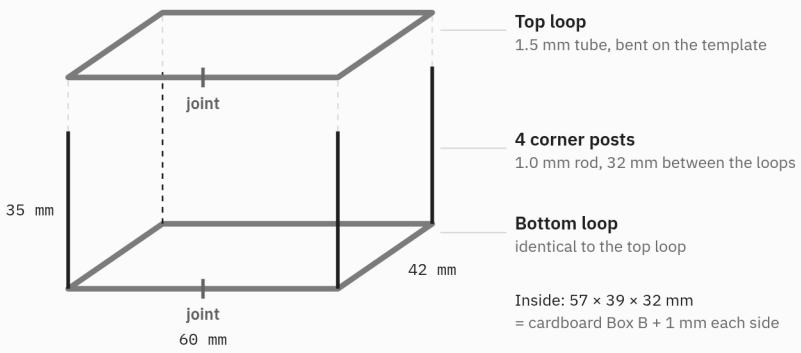
Two bent loops of 1.5 mm brass tube and four corner posts make the cage around the stacked layout. This session is all metalwork on an empty frame: no electronics anywhere near it. About three hours.

## Where you are

| 1–3 · done | 4a · done | 4b · now | 4c–4e · next |
|---|---|---|---|
| **It works** | **Mock-up** | **Brass frame** | **Finish and fit** |
| USB prototype, Wi-Fi, speaks at cap 45 | stacked layout, cardboard Box B | bend, cut, solder, wash the empty cage | insulate, mount, rewire, Wi-Fi check |

## The frame at a glance

Exploded view: the top loop drops onto the four posts.



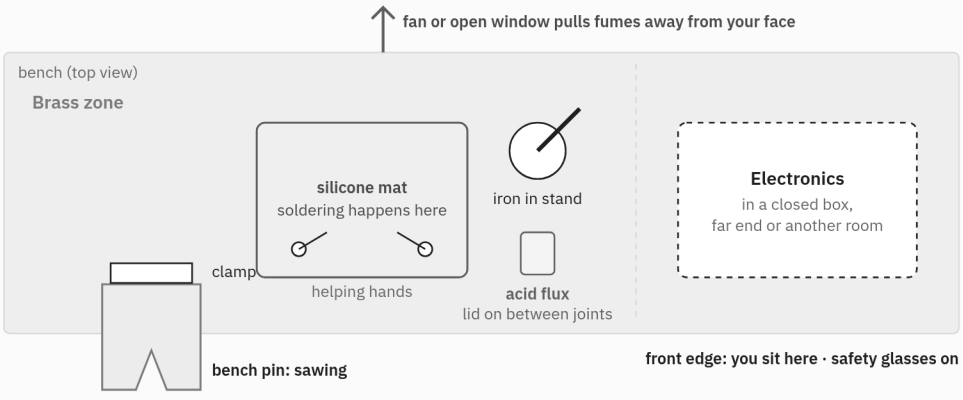

**Two identical loops, four posts.** The screen faces up through the top loop and A1 sits on the bottom layer. Corners are rounded in the real frame (3 mm bend radius); they are drawn square here for clarity. Both joints go on long sides, away from the corners.

## Before you start

1. **Print the template** `edu/pages/frame-template-letter.pdf` on cardstock: Letter, **Actual size**, black and white, one sheet at a time. Check its ruler measures exactly 100 mm.
2. **Buy baking soda and cotton swabs**. You need them right after soldering to neutralize the acid flux.
3. **Box up every electronic part**, including the spares and the loose prototype parts, and close the lid. The acid flux and brass chips must never reach them.

| PRODUCT YOU BOUGHT | USE | LINK |
|---|---|---|
| K&S #9831 brass tube, 1.5 mm OD × 0.225 mm wall, 4 tubes | two loops, one tube each | B005WPAW9M · caliper it: 1.5 mm |
| K&S #9861 brass rod, 1.0 mm, 5 rods | four posts and two joint pins | B005WPB7YG |
| WORKPRO jewelry pliers, 3-pack | bending (round-nose) | B0B8QBVXXR |
| SE 3-in-1 jeweler's saw, 144 blades, bench pin | every cut | B06XPSLS6N |
| BAISDY wet/dry sandpaper, 400–3000 | deburring | B07D2YYC11 |
| SE diamond needle files, 744DF-R | fitting the posts | owned · no link recorded |
| X-Tronic 3020-XTS station, tips, helping hands, silicone mat | soldering; keep one tip for brass only | owned · no link recorded |
| MAIYUM 63/37 rosin-core solder, 0.8 mm | the joints | B076QF1Y85 |
| Harris SCLF4 Stay-Clean acid flux | brass joints only | B0015DWPV8 |
| NEIKO digital caliper | checks | B000GSLKIW |
| 3M Solus 1000 safety glasses | sawing, filing, soldering | B016KZ1ZPM |
| Epic Medical Supply 91 % isopropyl alcohol | drying after the wash | B0BSJ93RKB |
| Amazon Basics cardstock, 65 lb, Letter | printing the frame template | B0DSH5V1TT |
| **Baking soda** (any plain sodium bicarbonate) | neutralizing the flux | **to buy** · e.g. 365 Baking Soda or any grocery store |
| **Cotton swabs** | scrubbing joints, the snag test | **to buy** · any pharmacy |

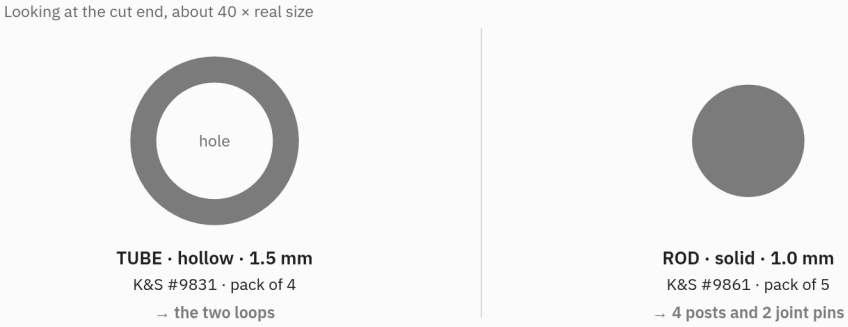
## 0:00 Set up the bench




**Two zones that never mix.** Zinc-chloride flux is corrosive and its fumes sting, so keep the jar closed, work with air moving away from you, and wash your hands after the session. Fit a spare iron tip and keep it for brass only: acid flux wears tips out.

## Tube or rod? Tell them apart first

Both come as 300 mm brass sticks and look alike. Look at the end of each stick and measure it before you start, then keep them in two separate piles.

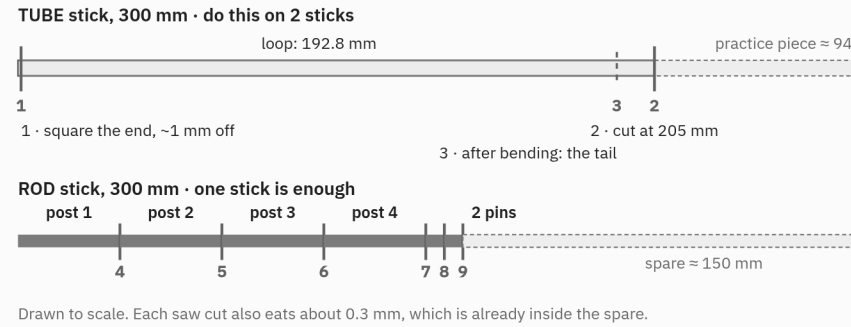



**See a hole in the end? It is the tube.** Confirm with the calipers: the tube reads 1.5 mm, the rod 1.0 mm. The tube listing covers several sizes, so if yours does not read 1.5 mm, stop and tell Claude before bending.

| STEP | USE | LINK |
|---|---|---|
| Mark the tube · Bend the corners | **Tube**, 1.5 mm, hollow · one stick per loop | B005WPAW9M |
| Close each loop: the hidden pin | **Rod**, 1.0 mm, solid · 2 × 6 mm | B005WPB7YG |
| Cut and fit the four posts | **Rod**, 1.0 mm, solid · 4 × 33 mm | B005WPB7YG |

## Every cut, in order

You saw at three moments: two quick cuts on each tube stick **before marking**, one cut on each loop **after bending**, and the rod pieces **after the loops are closed**. Nothing is cut to its final loop length while the tube is still straight.



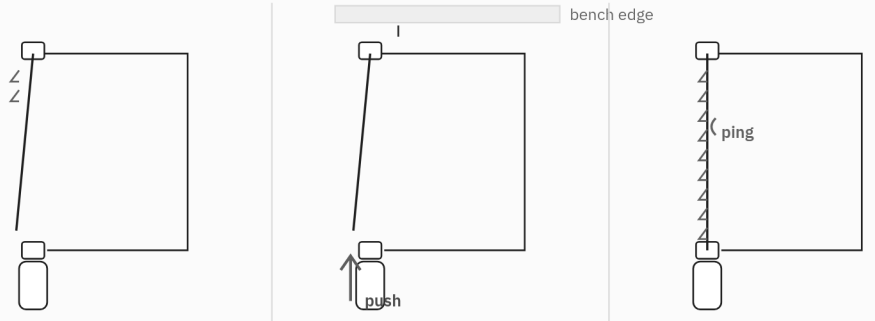

**Twelve cuts in total, in the order of the numbers.** Cuts 1, 2 and 3 happen on both tube sticks, so six tube cuts; cuts 4–9 are on one rod.

| CUT | WHEN | STOCK | WHERE | WHY |
|---|---|---|---|---|
| 1 | before marking | **TUBE** | ~1 mm off one end | a clean, square start: the “joint at 0” |
| 2 | before marking | **TUBE** | 205 mm from the squared end | loop + 10 mm tail + 2 mm spare; the rest is your practice piece |
| – | before bending | practice piece | a few cuts anywhere | learn the saw and test one bend on scrap |
| 3 | after bend 4 | **TUBE** loop | the tail, level with the start end | closes the loop exactly |
| 4–7 | loops closed | **ROD** | 4 × 33 mm | corner posts, filed to 32 later |
| 8–9 | loops closed | **ROD** | 2 × 6 mm | hidden joint pins |

## 0:10 Fit a blade and learn the saw

Tool: **SE 3-IN-1 JEWELER'S SAW** with its blades and V-slot bench pin · B06XPSLS6N · never flush cutters: they crush the 0.225 mm wall

1. **Pick a fine blade** from the 144-blade pack: size **2/0 or 3/0**, printed on each bundle. Finer blades cut thin tube cleanly; coarse ones catch and snap.
2. **Clamp the bench pin** to the table edge, V-slot facing you, flat side up.
3. **Fit the blade** as drawn below: top end first, then flex the frame, then the bottom end.

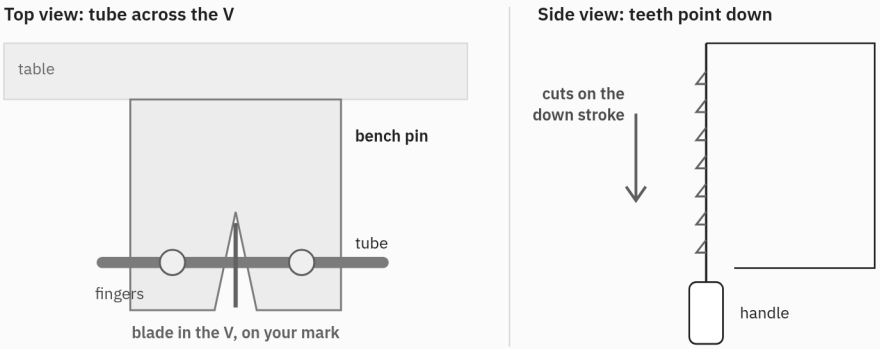



**1 · Clamp the top end**
teeth face out, point down

**2 · Flex the frame**
tip on the bench edge, push 1–2 mm

**3 · Clamp the bottom, let go**
blade is tight and pings

**Check the teeth with a finger:** slide it gently down the blade. Smooth going down and catching going up means the teeth point the right way. A slack blade wanders and snaps; a tight one cuts straight.




**The V supports the tube on both sides of the cut**, so the thin wall is not bent while you saw. Your fingers hold the stick down on either side of the V; the blade moves up and down inside it.

1. **Safety glasses on.** Sit so the bench pin is in front of your chest and you look straight down at the V.
2. **Place the stick** across the V with your mark centred over the gap. One finger presses it down on each side of the V.
3. **Hold the saw upright**, blade at 90° to the stick, resting on the mark.
4. **Start the cut** with two or three light *upward* strokes. They make a small notch so the blade does not skate off the mark.
5. **Saw with long, light strokes**, using most of the blade's length. It cuts on the down stroke; just guide it on the way up. If you have to push, the blade is dull or too coarse.
6. **Roll the stick toward you** a little every few strokes, so you cut around the tube instead of straight through it. The end stays round and square instead of pinched.
7. **Slow down at the end** and support the short piece with a finger so it does not snap off and bend.
8. **Deburr** with the 400 grit sandpaper (B07D2YYC11): rub the cut end flat on the sheet in small circles, then twist a scrap of sandpaper inside the tube's mouth. Drag a cotton swab across the end; repeat until it does not snag.

> **BLADES SNAP WHILE YOU LEARN**
>
> That is why the pack has 144. When one breaks, fit a new one the same way and carry on. A blade usually snaps from pushing too hard, twisting it in the cut, or a slack fit.

## 0:20 First cuts: square the end, cut to 205 mm

Use: **TUBE** · 1.5 mm, hollow · K&S #9831 · B005WPAW9M · two sticks, one per loop

1. **Cut 1, square the end.** Saw about 1 mm off one end of the stick. Factory ends are often pinched; this end becomes the loop's start, so it must be clean, round and square. Deburr

it.

2 **Mark 205 mm** from that squared end. Lay the stick on the template's **tube marking strip** (section 3) with the squared end on the 0 line and mark at the line labelled **cut 2 · 205**, just past the dashed tail. Your calipers only reach 150 mm, so use the printed strip for this one.

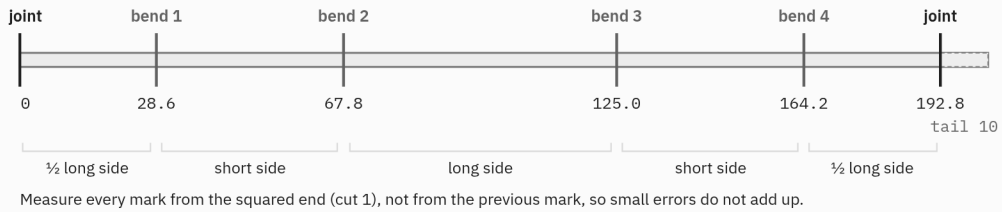
3 **Cut 2, at the 205 mark.** This is a forgiving cut: 2 mm of spare is built in, so it is a good one to learn on.

4 **Keep the short piece**, about 94 mm. It is your practice piece: saw two or three test cuts on it, and do one test bend on it before you bend the real stick.

5 **Repeat cuts 1 and 2 on the second stick.** Label the two long pieces “loop A” and “loop B” with the marker.

## 0:30 Mark the tube

Use: **TUBE** · 1.5 mm, hollow · K&S #9831 · B005WPAW9M



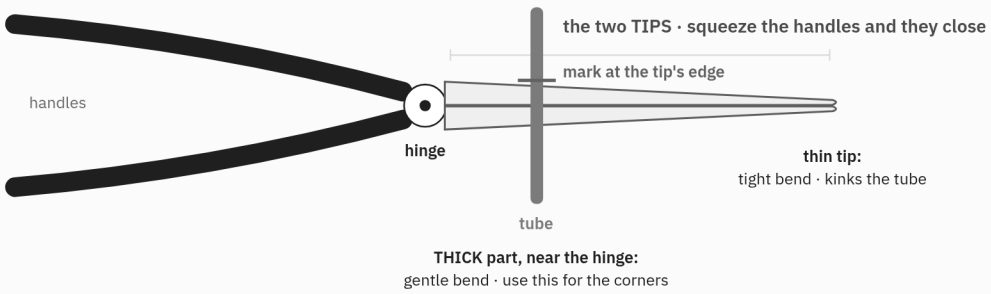

Measure every mark from the squared end (cut 1), not from the previous mark, so small errors do not add up.

**Lay the tube on the template's marking strip** and mark the four bend centres with a fine marker, all the way round the tube. Do this twice, one tube per loop. Leave the 10 mm tail on; it is cut off after bending.

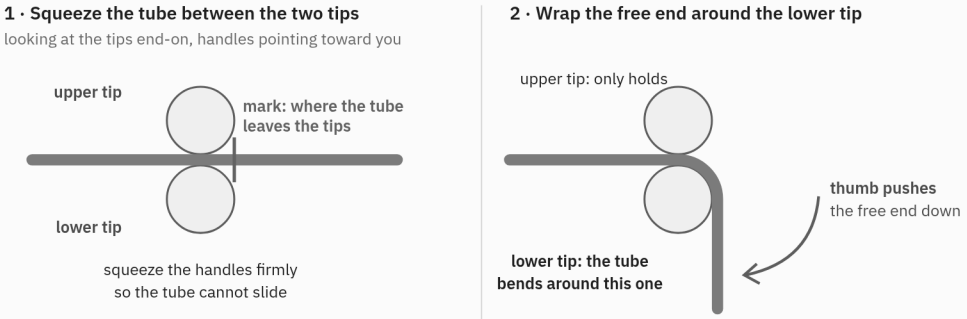
## 0:40 Bend the corners

Use: **TUBE** · 1.5 mm, hollow · the two sticks you marked

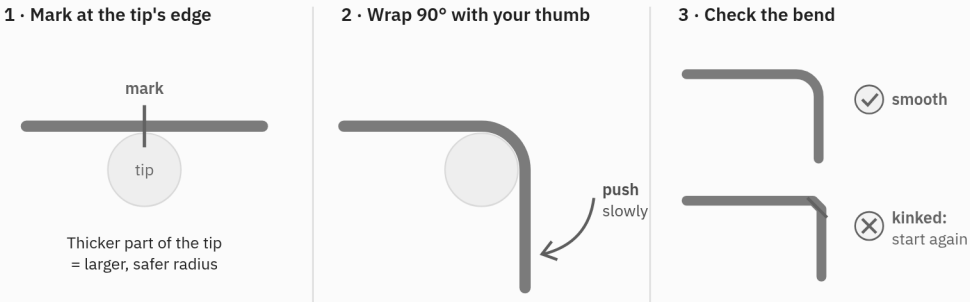
Tool: **ROUND-NOSE PLIERS** from the WORKPRO 3-pack · B0B8QBVXXR · the pair whose two tips are both smooth, round cones




**The tips are the two metal pieces at the front that close together when you squeeze the handles** (tool makers call them the jaws). On round-nose pliers each tip is a smooth cone: thick next to the hinge, thin at the tip. The thicker the spot you bend around, the larger and safer the corner. Hold the tube across the tips close to the hinge.




**The tube goes between the two tips, crossing them.** Both tips squeeze it so it cannot slide; then the free end is pushed around one tip only, the lower one here, until it points straight down: a 90° corner. Do this near the hinge, where the tips are thick.

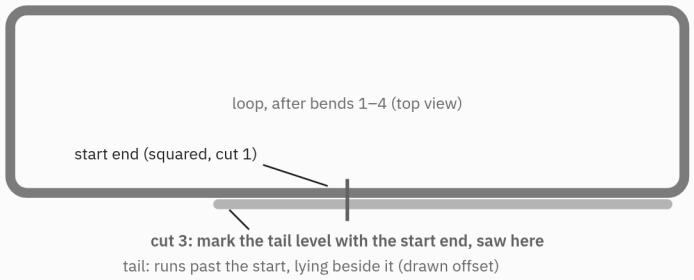



**Practise one bend on an offcut first.** The wall is only 0.225 mm thick, so a tight bend flattens it. After each corner, lay the tube on the template's grey band: the straight runs should sit on the band and the corner should follow its curve.

1. Bend 1 to bend 4 in order, all in the same direction, so the tube closes into a rectangle.
2. After bend 4, check the whole loop lies flat on the table. Twist gently by hand to correct it.
3. Repeat for the second loop. Lay the two loops on top of each other: they should match within about 0.5 mm.

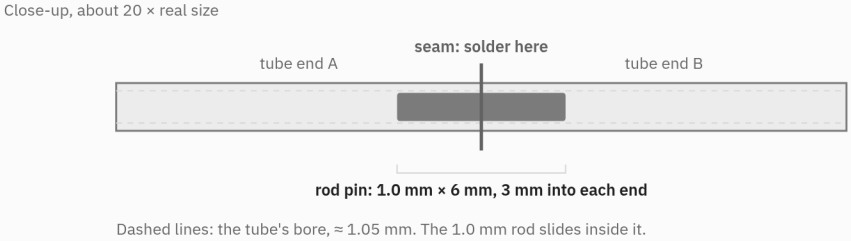
## 1:10 Close each loop

Use: the bent **TUBE** loops, plus **ROD** · 1.0 mm, solid · K&S #9861 · B005WPB7YG for the two 6 mm pins (cut later, in the posts step, or now if you prefer)




**Cut the tail only once all four bends are done.** Every bend shifts the length a little, so the true cut point is wherever the tail meets the start, not the 192.8 mark on the straight tube.

1. **Lay the loop flat.** After bend 4 the tail lies beside the start end.
2. **Mark the tail** exactly level with the start end's face. Check it is near your 192.8 mark; within 1–2 mm is normal.
3. **Spring the tail out** a few millimetres so only the tail sits across the bench pin's V, with the rest of the loop flat on the pin. Saw at the mark, rolling the tail as before.
4. **Square and deburr** the new end, then bring the two ends together: they should meet with no overlap and the loop should lie flat. A gap up to about 0.3 mm is fine; the pin and solder fill it. A bigger gap: close it by tightening the nearest corner slightly.
5. **Repeat for the second loop.**

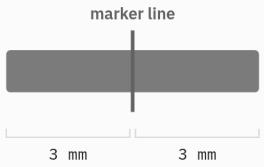



**A hidden pin keeps the joint straight and strong.** Push a 6 mm rod pin halfway into one end and slide the other end over it. You solder this seam in the soldering stage, together with the posts.

## Fitting the pin, step by step

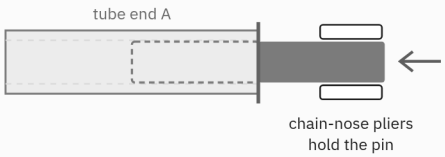
Use: the 2 short **ROD** pins (6 mm, solid 1.0 mm) · a fine marker · the second pair of WORKPRO pliers with flat, pointed tips (chain-nose) to hold the pin

**1 · Mark the middle of the pin**




Try it in a practice piece first: it should slide in with light pressure, not be hammered.

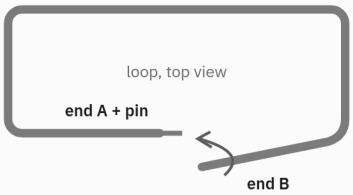
**2 · Push it into end A, up to the line**



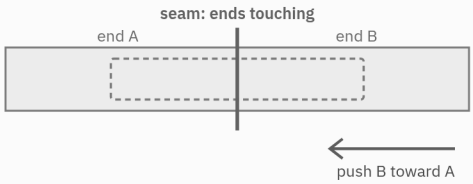

Push straight in until the line meets the tube's end. Tight? Twist gently. Never force it: the tube splits.

**3 · Spring the loop and line up end B**

The loop is springy: pull end B a few mm out of line, aim it straight at the pin, then let it come back.




**4 · Slide end B on until the ends touch**




The pin is now hidden inside (dashed). Only the seam shows; it is soldered later, with the posts.

**Pin first into one end, then the other end onto the pin.** Drawn about 30 × real size. The marker line on the pin tells you when it is halfway in, so each end gets 3 mm of pin.

**Check · no gap**



up to 0.3 mm is fine:
about 3 sheets of paper

bigger: tighten the nearest corner

**Check · straight at the seam**



a kink at the seam means the pin is not fully in one end: push it home

**Check · flat**



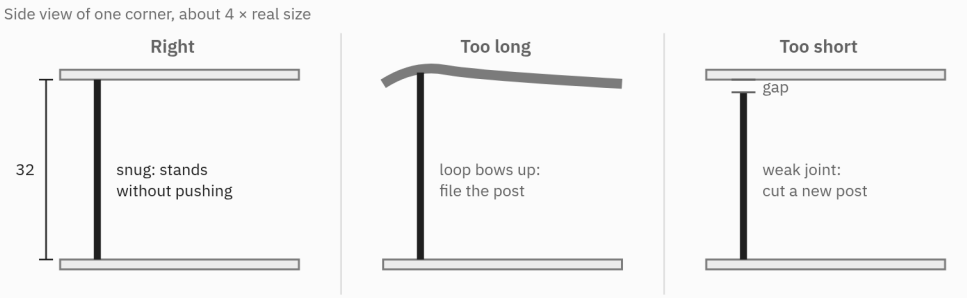
lay the loop on the table and press each corner: no rocking

**All three before you move on.** The pin is a friction fit, so the loop stays closed on its own until you solder it. If it keeps springing open, hold the seam in the helping hands when you solder.

> **IF THE PIN WILL NOT GO IN**
>
> Twist a small scrap of 400 grit sandpaper, or the finest round needle file, inside the tube's end to clear the burr, and try again. If it still will not go, leave the pin out and solder the two ends straight together, holding them in line in the helping hands.

## 1:25 Cut and fit the four posts

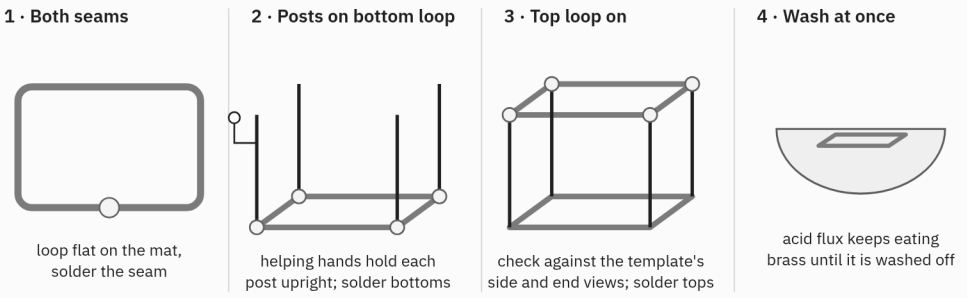
Use: **ROD** · 1.0 mm, solid · K&S #9861 · B005WPB7YG. Not the tube.



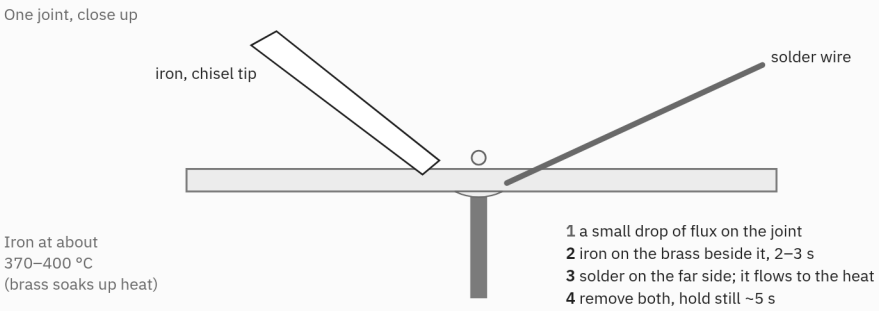

**Cut each post 33 mm, then file it down.** Cuts 4–7 in the cut plan. Check it standing between the two loops at its own corner and file a little at a time until it is snug. Number the posts and the corners 1 to 4 with the marker so each post goes back where it was fitted. Cut two 6 mm joint pins from the same rod.

1. **Square the rod's end** with a light cut, as on the tube.
2. **Mark all six pieces at once** on the template's **rod marking strip** (section 4): squared end on 0, then a line at each of cuts 4 to 9. Posts are spaced 33.5 mm and pins 6.5 mm; the extra 0.5 mm is what the saw removes.
3. **Saw on each line** on the bench pin, starting from the end. The rod is solid, so it takes a few more strokes than the tube; no need to roll it.
4. **The two pins come last** (cuts 8 and 9). Work over a sheet of white paper: short pieces fly off and are hard to find.
5. **File each post** to fit at its own corner, flat end, a little at a time, and number post and corner.

## 1:40 Solder the empty frame

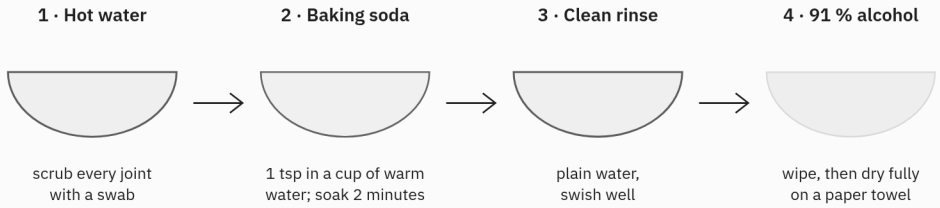



**All hot work happens on the empty frame, in this order.** Orange dots are the ten joints: two seams, four post bottoms, four post tops.



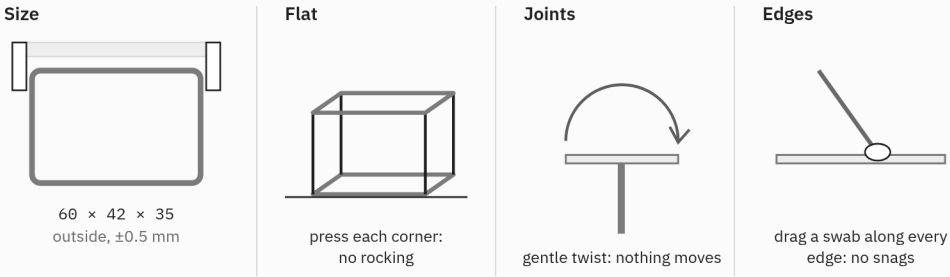

**Heat the metal, not the solder.** A good joint is a small, smooth fillet that wets both pieces. If solder balls up instead of flowing, the brass is not hot enough or needs a fresh touch of flux. Use as little solder as makes a fillet; blobs are hard to clean and show through paint.

## 2:20 Neutralize and wash




**Wash straight after the last joint.** Leftover zinc-chloride flux corrodes the joints and blisters paint later. If a joint still looks white or crusty after the wash, repeat steps 1 and 2.

## 2:35 Check the frame




**A snag on a swab is a burr that would later cut wire insulation.** Sand it until the swab slides over it cleanly.

- [ ] **FRAME-01 · THE EMPTY FRAME IS DONE**
  Outside 60 × 42 × 35 mm within about 0.5 mm, sits flat, all ten joints hold a gentle twist, no snags, washed and dry. Send Claude the three outside measurements.

## After the frame

> **ONE DECISION BEFORE ANY ELECTRONICS GO IN**
>
> The project rules still say the metal case stays off this USB device. Before mounting, Claude updates them with your OK: the frame is isolated and carries no current, it never touches ground, USB or either speaker wire, it is lined with Kapton wherever a part could touch it, and A1's antenna end stays as clear of brass as the layout allows.

| NEXT | WHAT IT INVOLVES |
|---|---|
| **Paint test (optional)** | spray the two whites on brass offcuts first; paint the frame only after the fit and Wi-Fi checks pass. Rust-Oleum satin white B000Z8FGII or gloss canvas white B000PIG3AS |
| **Dry fit** | the unpowered parts placed in the frame, nothing glued, to check the stack still fits |
| **Insulate and mount** | ELEGOO Kapton tape B072Z92QZ2 on module backs and wherever a wire crosses brass; the SHJADE low-temperature hot glue gun (owned, no link recorded) keeps modules in place |
| **Rewire and test** | CBAZY 30 AWG silicone wire B073RDGTPB, short soldered runs, one module at a time, re-checking each fast-track pass condition |
| **Wi-Fi check** | compare Wi-Fi on the open bench and inside the frame, several runs each |

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**Sources.** Frame size from the cardboard Box B net (55 × 37 × 30 mm inside) plus the builder's 1 mm margin; bend positions and post length from `edu/pages/frame-template-letter.pdf`. Tools and materials from `docs/INVENTORY.md`; product links are the listings on the original order sheet (`docs/MATERIALS.md`) for the brands the inventory records as bought, including the rod-in-tube fit and the acid-flux washing rule. Soldering temperature and the order of work follow the earlier frame notes in `docs/BUILD_GUIDE.md`, which is archived for the rest of the build.

**What is and is not verified.** Nothing on this page has been built yet. The frame size comes from cardboard estimates, not caliper measurements of the stacked parts, and the bend marks assume a 3 mm bend radius: your pliers may need a millimetre of correction, which is why the second loop is checked against the first. The checkbox is remembered only in this browser.